“Title – HOUSE PRICE PREDICTION ”

Project Report

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by

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ABSTRACT

The real estate market often requires accurate property price estimation to support buyers, sellers, and investors in making informed decisions. Manual estimation methods can be time-consuming and may lead to inconsistent results due to variations in location, property size, and amenities. This project, **AI House Price Predictor**, was developed to provide a simple and intelligent solution for estimating house prices using machine learning techniques.

The main objective of the project is to predict house prices based on important factors such as area, number of bedrooms, number of bathrooms, age of the house, and location. A sample dataset containing property details from cities such as Dehradun, Delhi, and Mumbai was used for model training and testing. Data preprocessing was performed using Pandas, including conversion of categorical location data into numerical format through one-hot encoding. Two machine learning algorithms, **Linear Regression** and **Random Forest Regressor**, were trained and evaluated using performance metrics such as **R^2 score** and **Root Mean Square Error (RMSE)**. The better-performing model was automatically selected for prediction.

A graphical user interface (GUI) was designed using Tkinter, allowing users to enter property details, compare the predicted price with their budget, and visualize price trends through graphs. The system successfully demonstrates how AI can simplify real estate price estimation with user-friendly interaction.

In conclusion, the project provides an efficient and practical house price prediction system. Future improvements may include using larger real-world datasets, additional features, and advanced algorithms to increase prediction accuracy.

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CHAPTER 1 INTRODUCTION

1.1 Topic Overview

Artificial Intelligence and Machine Learning are widely used for prediction tasks. House price prediction is one such application where property prices are estimated using factors like area, bedrooms, bathrooms, age, and location.

1.2 Background

House prices change depending on market demand, location, and facilities. Manual price estimation may not always be accurate. Machine learning provides faster and better predictions using past data.

1.3 Problem Statement

People often find it difficult to know the correct price of a house. Traditional methods are slow and may be inaccurate. Therefore, an automated prediction system is needed.

1.4 Aim and Objectives

The aim of this project is to develop an AI-based house price prediction system.

Objectives:

- Predict house prices using machine learning
- Compare Linear Regression and Random Forest models
- Select the best model
- Create a user-friendly GUI using Tkinter

1.5 Scope and Limitations

The project uses a small dataset and predicts prices based on limited features such as area, bedrooms, bathrooms, age, and location.

1.6 Significance

This system helps users estimate house prices quickly and supports better buying and selling decisions.

1.7 Method Summary

Dataset preprocessing, model training, testing, evaluation, and GUI-based prediction are used in this project.

CHAPTER 2

METHODS / MODELS / ALGORITHMS

2.1 Overview of Approach

This project uses Machine Learning techniques to predict house prices based on user inputs such as area, bedrooms, bathrooms, house age, and location. Two regression models were used and compared.

2.2 Models Used

(a) Linear Regression

Linear Regression predicts price by finding a linear relationship between input features and output price. It is simple, fast, and suitable for basic prediction tasks.

(b) Random Forest Regressor

Random Forest is an ensemble learning algorithm that uses multiple decision trees to improve prediction accuracy. It handles complex relationships better than Linear Regression.

2.3 Justification for Choosing Models

- **Linear Regression** was selected because it is easy to implement and understand.
- **Random Forest** was selected because it gives better accuracy for non-linear data.
- Both models were compared to choose the best-performing one.

2.4 Model Evaluation

The models were tested using:

- **R^2 Score** – Measures prediction accuracy
- **RMSE (Root Mean Square Error)** – Measures prediction error

The model with better performance was selected automatically.

2.5 Expected Behavior and Limitations

The system predicts house prices based on the given dataset. Accuracy depends on data quality and size. Since a small sample dataset is used, predictions may vary from real market prices.

CHAPTER 3 IMPLEMENTATION

3.1 Implementation Overview

The AI House Price Predictor was developed using Python. It predicts house prices using machine learning models and provides results through a graphical user interface.

3.2 Development Environment and Tools

- **Programming Language:** Python
- **Libraries Used:** Pandas, NumPy, Scikit-learn, Tkinter, Matplotlib, Joblib
- **Platform:** Windows / Python IDE

3.3 System Architecture and Modules

The system consists of the following modules:

1. **Dataset Module** – Stores house data such as area, bedrooms, bathrooms, age, location, and price.
2. **Preprocessing Module** – Converts location data into numerical form.
3. **Model Training Module** – Trains Linear Regression and Random Forest models.
4. **Prediction Module** – Predicts price based on user input.
5. **GUI Module** – Accepts input and displays results.

3.4 Key Implementation Details

- Dataset is loaded into a Pandas DataFrame.
- Train-test split is used for model testing.
- Best model is selected using R^2 score.

3.5 Data Handling

The dataset contains features like area, bedrooms, bathrooms, age, and location. Dummy variables are used for location names.

3.6 User Interface Implementation

Tkinter GUI was created with input boxes, location dropdown menu, prediction button, graph button, and result display section.

3.7 Challenges and Resolution

Small dataset size was a challenge. This was managed by comparing two models and selecting the best one automatically.

CHAPTER 4

OUTPUT / KEY RESULTS

4.1 Introduction

This chapter presents the results obtained from the AI House Price Predictor system. The outputs include model performance, predicted house prices, and graphical analysis.

4.2 Quantitative Results

Two machine learning models were tested:

- **Linear Regression**
- **Random Forest Regressor**

The model with higher **R^2 Score** and lower **RMSE** was selected as the best model for prediction.

4.3 Prediction Output

The system accepts user inputs such as area, bedrooms, bathrooms, age, location, and budget, then displays:

- Predicted House Price
- Budget Status (Within Budget / Budget Too Low)
- Best Model Name
- Accuracy Score
- Error Value

4.4 Comparative Results

Random Forest performed better than Linear Regression on the sample dataset, so it was selected automatically.

4.5 Graphical Results

A scatter graph of **House Price vs Area** was generated using Matplotlib.

Interpretation:
The graph shows that house prices generally increase with increase in area.

4.6 Summary of Results

The system successfully predicted house prices and provided quick, user-friendly results through the GUI.

CHAPTER 5

LIMITATIONS AND FUTURE ENHANCEMENTS

5.1 Limitations

1. The project uses a **small sample dataset**, so prediction accuracy may be limited.
2. Only a few features are considered such as area, bedrooms, bathrooms, age, and location.
3. Real estate market conditions change frequently, but live market data is not used.
4. Predictions are limited to selected cities only.
5. Accuracy depends on the quality of training data.

5.2 Future Enhancements

1. Use a **large real-world dataset** for better accuracy.
2. Add more features such as parking, floor number, nearby facilities, and furnishing status.
3. Integrate live market data from property websites.
4. Expand prediction for more cities and locations.
5. Use advanced machine learning or deep learning models.
6. Develop a web or mobile application for wider user access.

CHAPTER 6

CONCLUSION

The project **AI House Price Predictor** was developed to solve the problem of estimating house prices quickly and accurately using machine learning techniques. The main objective was to predict house prices based on features such as area, bedrooms, bathrooms, age, and location through a user-friendly system.

The project successfully implemented **Linear Regression** and **Random Forest Regressor** models, compared their performance, and selected the best model for prediction. A GUI was also created using Tkinter to make the system simple and interactive for users.

Although the project uses a small dataset and limited features, it demonstrates the practical use of Artificial Intelligence in the real estate field. In future, larger datasets, more features, and advanced models can improve accuracy further.

Overall, this project shows how machine learning can simplify property price estimation and support better decision-making in the housing market.

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